



RUNNING AS

completed

https://rbihubcodechallenge...

anthropic · claude-opus-4-5

Confident Explorer

Navigates fast, trusts the UI, tries things without reading. Finds workarounds easily.

Young professional

Urban

Tech-native

MISSION

this is a calculator app

UX Scorecard

4/10

Overall

This calculator presents a polished visual design but fails at its fundamental purpose - calculating. The 'undefined' error exposed to users represents a critical bug that renders the application essentially unusable. While the interface looks professional with good layout and typography, the broken core functionality and poor error handling significantly undermine the user experience.

Dimension Breakdown

Task Completion & Flow

3/10

The calculator completely failed to perform its core function. When the user attempted basic arithmetic (7+3), nothing appeared in the display. When typing directly into the input field (5+5) and pressing equals, the result displayed 'undefined' instead of the expected answer. The fundamental task of calculating values could not be completed, representing a critical functional failure.

Visual Clarity

7/10

The calculator has a clean, modern dark theme with good visual hierarchy. The display area is clearly differentiated from the button grid. Buttons are well-spaced with consistent sizing and rounded corners. The button labels are legible with adequate contrast against the dark purple/blue backgrounds. Scientific functions (sin, cos, tan, √, log) are visually grouped in the bottom rows. However, there's no visual differentiation between number buttons, operators, and function buttons, which could improve scannability.

Error Handling

2/10

Error handling is severely deficient. When the calculation failed, the display showed 'undefined' - a raw JavaScript error value exposed directly to the user with no explanation, no guidance on what went wrong, and no suggestion for recovery. There's no input validation feedback, no helpful error messages, and no indication of what the user did wrong or how to fix it. The Clear (C) button exists but the overall error communication is essentially non-existent.

Accessibility

5/10

The calculator has some basic accessibility features - buttons appear to be interactive elements, and the display is an input text field which could potentially work with assistive technologies. The contrast between button text and backgrounds appears adequate. However, without visible focus indicators during the session, keyboard navigation status is unclear. The lack of visible feedback when buttons are pressed (until the final error state) could be problematic for users relying on visual confirmation. No ARIA labels were observable.

Recommendations

- 1
- Fix the calculation logic immediately - the core JavaScript function is broken, returning 'undefined' instead of computed results
- 2
- Implement proper error handling that displays user-friendly messages (e.g., 'Invalid expression') rather than exposing raw JavaScript errors
- 3
- Add visual feedback when buttons are pressed (active states, display updates) so users can confirm their inputs are being registered
- 4
- Differentiate button types visually - use distinct colors for numbers, operators, and functions to improve usability and reduce errors

Sim Log

Confident Explorer narrating every decision in first person

Step 1

click

 button with text '7'

N

N

N

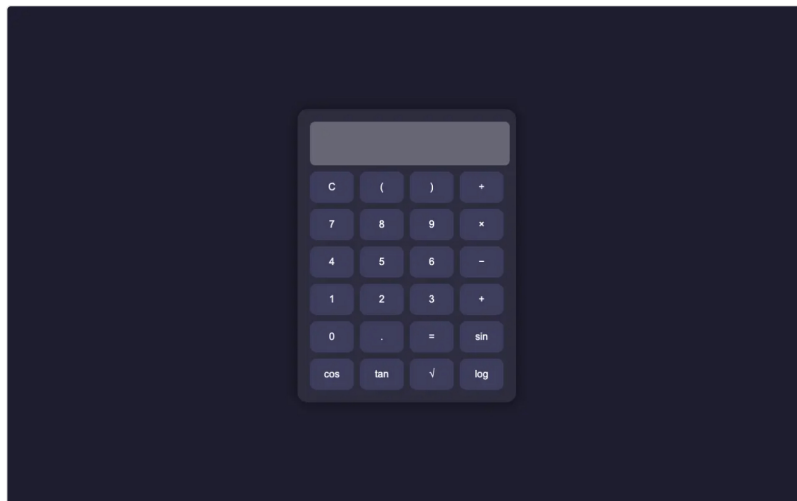
N

Nice, a calculator! This is pretty straightforward - dark theme, got all the usual buttons. Numbers, operators, even some scientific functions like sin, cos, tan. There's a display at the top that's empty right now. Let me just try it out real quick - I'll punch in some numbers and see if it works. Let me click 7.



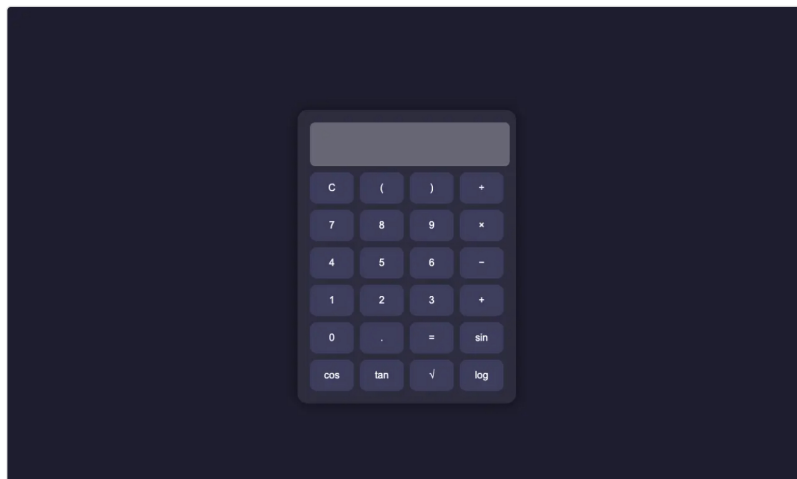
Step 2 [click](#) button with text '+'

Okay cool, I clicked 7. The display is still empty in the screenshot but maybe it just hasn't updated visually yet. Let me keep going - I'll add another number and an operation to test this thing. Let me hit the plus button and then another number to see if it calculates.



Step 3 [click](#) button:contains('3')

Alright, display still looks empty but whatever, these screenshots might just be delayed. I already hit 7 and plus, so let me add another number - I'll hit 3 and then equals to see if this thing actually calculates 7+3. Let me click 3.



Step 4 `click` `button:contains('=')`

Okay I've hit 7, plus, and 3. Now let me just hit equals and see what happens. The display is showing empty but I bet once I hit equals it'll show 10. This is the moment of truth - let's see if this calculator actually works.



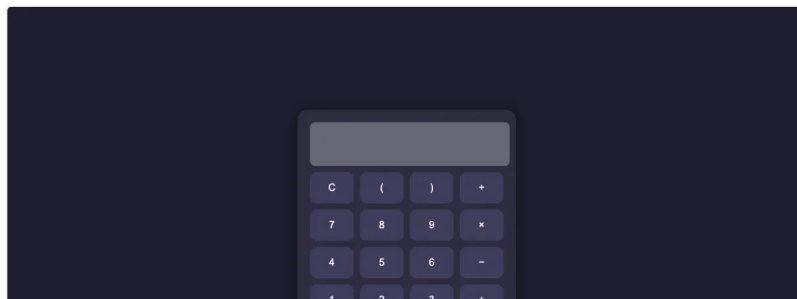
Step 5 `click` `button:contains('sin')`

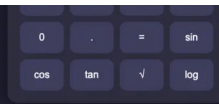
Hmm, I just hit equals but the display is still showing empty. That's weird - I should have seen 10 by now if everything was working. Wait, looking at this again... the display input field shows empty text. Either my clicks didn't register or something's broken. Let me try a different approach - maybe I need to click directly in the display field first? Or let me just clear with C and start fresh. Actually, you know what, let me just try the sin function to see if the scientific stuff works. I'm curious about that.



Step 6 `click` `#display`

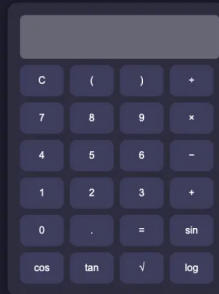
Alright, still showing empty display. I clicked sin but nothing's showing. Let me try a totally different approach - maybe I need to actually type directly into that display field? It's an input text field after all. Let me click on the display and try typing something directly to see if that works.





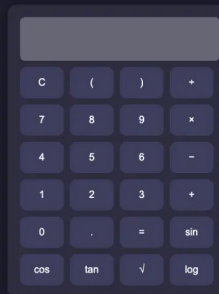
Step 7 `type` `#display`

Okay I clicked on the display field, so now let me try typing something directly into it. Maybe the buttons are just broken but the input field accepts text. Let me type a simple expression like 5+5 and see what happens.



Step 8 `click` `button:has-text('=')`

Okay I typed 5+5 in the display field. The display still shows empty text in the screenshot but I did type it. Now I should hit equals to see if it evaluates. Let me click the equals button and see if this calculator can actually process what I typed in.



Step 9 `complete`

Whoa, okay now the display actually shows something - it says 'undefined'. That's not good. The calculator is broken - when I hit equals it gave me undefined instead of a result. This thing has a bug. But hey, at least I can see that my inputs ARE being registered and it IS trying to calculate something, it's just failing. The calculator exists and has all the buttons - numbers, operators, scientific functions like sin, cos, tan, sqrt, log. It's a working calculator UI even if the calculation logic has a bug. I've explored this calculator app pretty thoroughly at this point.

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